GW-1
SOLAR // WRAITH

POCKET GAMMAWAVE · CONTROLS

every button, knob, layer, combo and page

firmware v1.2.1-GW · M-VAVE FM-1 · gammawaveaudio.com · shadowtune.itch.io/gammawave · fvalente.github.io/GammaWave-FM1

THE PANEL

KNOBS AND TRANSPORT

MASTER	Master volume.
SELECT	**Scrolls the section's pages** (HOME: the 49-genre generator). Held with a button, its **dice** (OK keeps, CANCEL back): HOME the generator, EDIT the sound, SEQ the sequence, LFO the LFO, FX the effects. GLO + SELECT: the tempo. In the generator: its pages, the reel, the SET (round).
PRESETS	Browses the sounds (factory and user). On step, list and song pages: the cursor / the row. In the generator: NOW and the **TAKES** you saved (scroll, round: a take plays on the next bar; **OCT+ held: the keys shape NOW**). **Hold a page button + PRESETS: jump through its pages.**
ALGORITHM	Selects the track from any page: T1–T4, or T1–T6 with TRACKS 6. In the generator: new instruments for the piece (the notes kept).
KNOB 1–4	The four columns on screen. Fast sweeps the whole range, slow fine-tunes. With a layer held: the layer's.
OCT– · OCT+	The keyboard's octave (both together: back to 0). On action pages and dialogs: back / confirm.
PLAY	Start / stop. GLO + PLAY: restart from the top.
REC	Arms live recording on the selected track (notes and knob moves, overdub).
HOME	Tap: the ecosystem. **Hold: the MENU.**
KEYS	27 keys play the selected track in its scale; with a layer held they become that layer's.

PAGE BUTTONS

BUTTON	TAP: ITS PAGES, ONE PER TAP	HOLD
ENV	ENV · ENV DEST	–
LFO	LFO · LFO DEST · MOD · MSEQ · MSEQ 2	MACROS M1–M4 (stay latched)
FX	TRACK · SLICER · DELAY · REVERB · CHORUS · MASTER (two quick taps: next section)	PUNCH-IN FX
SCL	SCL · SCL 2 · CHORD	ROOT by key
EDIT	EDIT 1 · EDIT 2 · FILTER · VOICE (FM6: OPERATOR · OP ENV · OPERATOR 2; DRUM: KIT · KIT 2 · KIT 3 · X-MOD · FILTER · SOUND 1–4 · LEVELS 1–2 · STEREO)	INIT · RECALL · sound mutes
GLO	MIXER · GLOBAL · SYSTEM	MUTE · SOLO · LEVEL · TAP
SAVE	PRESETS · USER · PROJECT · ARRANGE · TOOLS	SONG layer
ARP	ARP · ARP 2	–
SEQ	STEP · NOTE · TIMING · STRETCH · COMP · COMP 2 · WHEN · PATTERN · SHIFT · HUMAN · PLAYHEADS · AUTOMATION · AUTO LIST · MUTATE	SEQ TOOLS
REC	(transport)	CLEAR · CLICK

BUTTON + PRESETS jumps through its pages (no wrap). SELECT alone scrolls the section's pages.

LAYERS · HOLD A BUTTON

FX · PUNCH-IN EFFECTS

F#	G#	A#	C#	D#	F#	G#	A#	C#	D#	F#
M1	M2	M3	M4	M5	M6		TGT			

F3	G3	A3	B3	C4	D4	E4	F4	G4	A4	B4	C5	D5	E5	F5	G5
REP 1/8	REP 1/16	REP 1/32	LPF	HPF	OCT–	OCT+	½ TEMPO	DEC–	DEC+	STUT 1/8	STUT 1/16	ATK+	ARP	RAND	ERASE

A key plays its effect while held (MENU FX LATCH ON: toggles); hold FX and let go untouched: **LOCKED** open, FX again unlocks; OCT- all off. In on the next 1/16. Black 1–6 mute T1–T6, A#4 picks which tracks the MIDI ones touch (all · synths · drums). With REC armed and playing they record into the section's lane; G5 erases it.

KNOBS

FILTER	CRUSH	THROW	DEPTH

GLO · MIX

F#	G#	A#	C#	D#	F#	G#	A#	C#	D#	F#
M1	M2	M3	M4	M5	M6	KICK	SNR	HAT	PERC	

F3	G3	A3	B3	C4	D4	E4	F4	G4	A4	B4	C5	D5	E5	F5	G5
SOLO 1	SOLO 2	SOLO 3	SOLO 4	SOLO 5	SOLO 6	ALL	TAP	UNDO							

Black 1–6 mute T1–T6 (latched), 7–10 the DRUM-X groups. White: solo while held, E4 unmutes all, F4 tap tempo, G4 **UNDO** the last big change (again: redo), A4 **FREEZE** MUTATE. SELECT: the tempo. PLAY: restart.

KNOBS

LEVEL T1	LEVEL T2	LEVEL T3	LEVEL T4

T5 or T6 selected: the knobs are T3–T6.

SAVE · SONG

F#	G#	A#	C#	D#	F#	G#	A#	C#	D#	F#

F3	G3	A3	B3	C4	D4	E4	F4	G4	A4	B4	C5	D5	E5	F5	G5
PLAY A	PLAY B	PLAY C	PLAY D	STORE A	STORE B	STORE C	STORE D					LOOP/ SONG	SONG REC		SONG

PLAY A–D on the next bar (several in one hold: a quick chain), STORE the section into A–D. OCT+ stores the playing section, OCT– recalls it. On the SONG page KNOB 4 stores a scene.

KNOBS

SONG 1–8	-	-	-

EDIT · SOUND

F#	G#	A#	C#	D#	F#	G#	A#	C#	D#	F#
BD	SD	CP	CH	OH	TM	RS	CB			

F3	G3	A3	B3	C4	D4	E4	F4	G4	A4	B4	C5	D5	E5	F5	G5
INIT	RECAL														

INIT: the factory sound (DRUM: the kit too). RECALL: the sound stored in the section. On a DRUM track black 1–8 mute each DRUM-X sound.

KNOBS

EDIT 1	EDIT 2	EDIT 3	EDIT 4

GEN · OCT+ HELD: SHAPE NOW

F#	G#	A#	C#	D#	F#	G#	A#	C#	D#	F#
KIT	BASS SND	PAD SND	LEAD SND	TEX SND	ARP SND	ARR –	ARR +	ORIG	JAM	CODE

F3	G3	A3	B3	C4	D4	E4	F4	G4	A4	B4	C5	D5	E5	F5	G5
DRUM	BASS	CHRD	ARP	LEAD	TEX	ALL SND	HARM	DROP DRUM	DROP BASS	DROP CHRD	DROP ARP	DROP LEAD	DROP TEX	MIX	VARY

GEN (OCT– + PLAY), any page but the SET's. The piece playing is **NOW**: hold **OCT+** and the keys are silent and change NOW, on the next bar (stopped: at once) — **no take** (SAVE keeps NOW as one). The whole screen maps the keys in their parts' colours and says the key down in full. Left half: white F3–D4, let go: that part's notes made again (KNOB 4 on a section: there only); the black key beside it: its **sound**. Right half, mirrored: G4–E5 **drop** the same parts while held. **Hold a part or sound key ½ s: lock it**. On a take's card: the key **borrows** its part. C#5 **ORIG**: back to the take NOW came from. D#5 **JAM**: NOW changes by itself (held ½ s: every 4, 8, 16, 32 bars). F#5: the **share code**. Held to read the map, let go: nothing.

KNOBS ON THE REEL

TEMPO	KEY	ENERGY	SECTION

GEN · NOW, THE TAKES · OCT– HELD: PLAY THEM

F#	G#	A#	C#	D#	F#	G#	A#	C#	D#	F#

F3	G3	A3	B3	C4	D4	E4	F4	G4	A4	B4	C5	D5	E5	F5	G5
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16

Hold **OCT–**: the white keys play **takes 1–16** on the next bar (stopped: at once), like clips; the screen shows the one playing and the next. A tap of OCT– still goes back.

SAVE	**NOW kept as a take** (the next number). OCT+ held + SAVE: type a **share code**.
PRESETS	NOW, then the takes, round: a take plays on the next bar. Stopped: OCT+ or PLAY.
K1 · K2 · K3	**TEMPO** at once; **KEY**, **ENERGY**: NOW made again once the knob rests.
ALGORITHM	New instruments for NOW, the notes kept.
GLO + G4 · A4	**UNDO** NOW a step (8) · forward. At its start: the music before GEN.
EDIT	Delete the take (OCT+ yes). Held: every take.
REC	Playing: record a **SET** of the takes you play.
SELECT	The SET page: K1 entry (playing: it next bar), K2 take, K3 bars, PLAY.

NOW's card says TAKE n or EDITED and where it came from; a take's card its ID, its parent and what changed. What a control did shows in a band at the foot of the screen. The takes (32), the SET and JAM are kept over power-off.

LFO · MACROS

Knobs M1–M4; they stay latched when let go (LFO again: off). F3 rolls new routes, G3 clears them (only tracks that sound). LFO + SELECT: the track's LFO diced.

KNOBS

M1	M2	M3	M4

SCL · SCALE

Any key sets the ROOT. ROOT and SCALE are every melodic track's (and RESO's).

KNOBS

ROOT	SCL	OFS	PIT

SEQ · TOOLS

On the SEQ pages: F3 CLEAR · G3 REVERSE · A3 B3 SHIFT · C4 RANDOM · D4 COOK · E4 BEAT · F4 CLR AUTO (the automation).

KNOBS

LEN	DIV	SWING	GATE

REC · RECORD

F3 CLEAR the track (at once; GLO + G4 undoes it), G3 the metronome CLICK.

KNOBS

CLICK	-	-	-

COMBOS & [OCT–] SHIFT DECK

GAMMAWAVE [OCT–] SHIFT SHORTCUTS & BUTTON COMBOS

OCT– + ARP	**HEARTH MACROS (1/6–6/6)**: Gravity, Singularity, Photosphere, Wraith Scream, Formant Vowel, Solar Eclipse — drive type and amount, cutoff, resonance (4–6: SCREAM / VOWEL filter). Undo: OCT– + GLO.
OCT– + PLAY	**49-GENRE SONG GENERATOR (GEN)**: Trap ATL, UK Drill, Dark Trap, Rage Trap, house, techno, IDM, ambient… KNOB 1 genre, KNOB 2 seed, KNOB 3 energy, KNOB 4 form; SELECT pages (parts, SONG: tempo key scale length, FEEL: space fills wet grit), ALGORITHM new instruments, the piece playing is NOW (every change shapes it, no take), SAVE keeps it as a take; PRESETS the TAKES reel (kept over power-off: scroll to play one, K1 tempo, K2 key, K3 energy, K4 a section, EDIT deletes; GLO + G4 / A4 NOW back / forward; hold OCT+: F3..D4 a part made again, black keys its sound, G4..E5 drop a part while held, C#5 back to the take, D#5 JAM, F#5 the share code; hold OCT–: the white keys play takes 1–16; REC a SET); OCT+ makes, OCT– back, PLAY auditions.
OCT– + ENV / SCL	**SEQ > MUTATE (1-CLICK)**: jumps to MUTATE; if it was off, switches it on (Sound 2 bars, Pattern 4 bars, Amount 50%).
OCT– + HOME / SAVE	**HYPERSPACE VISUALIZER**: 4 animated scenes (Supernova, Wraith 3D, Warp Tunnel, Matrix FFT); OCT+ next scene, OCT– leaves.
OCT– + SEQ · EDIT · FX · LFO	**1-HAND DICE**: rolls Sequence, Sound Patch, Track FX, or LFO on the active track.
OCT– + GLO	**UNDO**, again: REDO (same as GLO + G4).
HOME · EDIT · SEQ · LFO · FX + SELECT	HOME: the 49-genre generator (GEN). The others dice: sound · sequence · LFO · effects (KEEP? OK / CANCEL).
PAGE BUTTON + PRESETS	Jump through that button's pages.
REC + FX EDIT ENV LFO SCL ARP GLO	Clear that part's automation on the selected track (FX: the punch-in lane too).
SEQ + REC	CLEAR ALL SEQUENCES? (OCT+ confirms).
SAVE + OCT+ / OCT–	Store / recall the playing section.
GLO + PLAY	Restart from the top.
GLO + SELECT	Tempo (from any page).
STEP HELD + KNOB	A parameter lock on that step (roll, grid).
HOME held	MENU: knob accel, FX latch, hold time, USB level, palettes, animation…

EVERY PAGE · KNOB 1 – 4

ENV

ENV	ATK	DEC	SUS	REL
ENV DEST	FILTER	PITCH	SHAPE	LOOP

LFO

LFO	RATE	WAVE	PHASE	FADE
LFO DEST	PITCH	FILTER	SHAPE	AMP
MOD	SLOT	SRC	DEST	AMT
MSEQ	STEP	LEVEL	LEN	SLEW
MSEQ 2	DIV	-	-	-

SCL · ARP

SCL	ROOT	SCALE	QUANT	TRANS
SCL 2	ALL TR - - -			
CHORD	CHORD	VOICING	STRUM	-
ARP	MODE	RATE	OCT	GATE
ARP 2	SWING	PROB	HOLD	ORDER

FX

FX	DIST	CHORUS	DLY	REV
DIST	DIST	TYPE	TONE	-
AUTO PAN	DEPTH	RATE	SYNC	WAVE
SLICER	MODE	PAT	RATE	DEPTH
SL BUS	T1	T2	T3	T4
SL BUS 2	T5	T6	FX	-
SL PITCH	STEP	PITCH	LEN	QUANT
DLY	TIME	FEEDBK	TONE	MIX
DLY 2	WIDE	PITCH	SPRAY	MOD
DLY 3	TYPE	PING	-	-
REVERB	TYPE	SIZE	DAMP	PRE/AIR
REVERB 2	MOD	RATE	FILT	WIDE
CHORUS	RATE	DEPTH	STEREO	-
MASTER	CLIP	PUNCH	DUCK	REL

EDIT

EDIT 1	E1	E2	E3	E4
EDIT 2	E5	E6	E7	E8
RETRIG	RTRG	-	-	-
OPERATOR	OP	RATIO	FINE	LEVEL
OP ENV	OP	STAGE	RATE	LEVEL
OPERATOR 2	OP	MODE	DETUNE	VEL
FILTER	TYPE	CUT	RES	DRIVE
VOICE	VOICE	GLIDE	MODE	PRIO
KIT 3	BP	FEEDBK	-	-
X-MOD	RATE	DEPTH	SHAPE	RAND
LEVELS	KICK	SNARE	CLAP	HATCL
LEVELS 2	HATOP	TOM	RIM	BELL
SOUND	SOUND	SIDE	PITCH	PMOD
SOUND 2	SOUND	DECAY	NOISE	COLOR
SOUND 3	SOUND	MODE	PM.TYPE	DRIVE
SOUND 4	SOUND	LAYER	FEEDBK	SEND
STEREO	IMAGE	-	-	-

SEQ

STEP	STEP	NOTE	TIME	FLAG
STEP · DRUM	STEP	LANE	HIT	ACC
COMP	STEP	TYPE	VAL	VAL
COMP 2	STEP	TYPE	VAL	VAL
WHEN	STEP	A	B	LEN
PATTERN	LEN	DIV	SWING	GATE
AUTOMATION	PLAY	EVENT	LOCK	CLEAR
AUTO LIST	ROW	STEP	PARAM	VALUE
MUTATE	SOUND	PATTERN	AMOUNT	MODE
PLAYHEADS	HEAD	RATE	DIR	INTVL
NOTE	FOCUS	PITCH	MOVE	LEN
TIMING	STEP OFFSET REC Q -			
STRETCH	x/÷	REPEAT	NUDGE	LEN
HUMAN	VEL	TIME	GATE	-
SHIFT	OFFSET	PITCH	CHANCE	C.LEN

GLO · SAVE

MIXER	LEVEL	PAN	REV	MUTE
GLOBAL	BPM	SWING	CLOCK	TUNE
SYSTEM	TRACKS	LOAD	ROUTE	CPU
PRESETS	No.	ENG	FAV	LIST
USER	SLOT	LOAD	ERASE	SAVE
PROJECT	PROJ	SECT	LOAD	SAVE
ARRANGE	ROW	SECT	BARS	SCENE
TOOLS	CLEAR SEQ · INIT SOUND · DEL ROW · CLEAR SONG · NEW PROJECT · CPU TEST: K1 picks, OCT+			

GPL-3.0 · Companion to GammaWave Desktop VST (gammawaveaudio.com) · Built on Felucca, JIANT & SLOOP